

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement 3.3V PCAP Monte Carlo SPICE Model Document

**Version 0.1 Phase 2
2010/12/27**

Table of Contents

1	INTRODUCTION	5
1.1	Revision History	5
1.2	General Information.....	5
1.3	Model Usage	5
1.3.1	Simulators	5
1.3.2	The combination of statistical model & corner model	6
1.3.3	Sigma number for Monte-Carlo model.....	6
1.3.4	Model usage.....	6
2	MONTE CARLO SIMULATION RESULTS FOR PROCESS AND MISMATCH VARIATION	10
2.1	Simulation Result Plot For Process Variation	10
2.2	Simulation Result Plot For Mismatching Variation	11
3	APPENDIX.....	12
3.1	Hspice Warning Message	12
3.2	Eldo Warning Message	12
3.3	Spectre Warning Message	12

List of Figures

Figure 1-1 PCAP layout.....9

Figure 2-1 Monte Carlo plot of PCAP's capacitance10

Figure 2-2 Mismatching simulation of PCAP's capacitance11

List of Tables

Table 1-1 Revision history5

Table 3-1 Hspice warning message12

Table 3-2 Eldo warning message12

Table 3-3 Spectre warning message12

1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p2	2010/12/27	Po Shun Chang	1.Original release. 2.This Monte Carlo model includes both process and mis-matching variation characteristics extracted from model cards as below : l110ae_pcap33_v012.mdl l110ae_pcap33_v012.lib

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI Advanced Enhancement Process

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.11_1.1

1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. The user can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo PCAP model
typ: Typical PCAP case
max: Maximum PCAP case
min: Minimum PCAP case

1.3.3 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

1.3.4 Model usage

For Synopsys Hspice

- Copy model files into your simulation directory.
- Define sigma value for MC simulation.

Example: .param sigma=3

.lib "/1110ae_pcap33_v012_mc_corner.lib" CaseName

, where CaseName could be mc, typ, max, or min.

- Define device condition by 'Xxx' statement.

Example:

Xncap Vgg Vbb DeviceName lf=2u wf=5u nf=4 mf=1 m=1 mis_flag=1

- m: number of devices

- mf: the same as "m"

-mis_flag: the flag parameter is used to turn on/off a mismatching parameter for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

- Define the control of Monte-Carlo simulation.

Indicator:

PROCESS: 1 to turn on the process MC simulation
 0 to turn off the process MC simulation

MISMATCH: 1 to turn on the mismatch MC simulation
 0 to turn off the mismatch MC simulation

Example 1:

.PARAM PROCESS=1 MISMATCH=1 * turn on process and mismatch MC simulations at the same time.

Example 2:

.PARAM PROCESS=1 MISMATCH=0 * just turn on the process MC simulation only.

Example 3:

.PARAM PROCESS=0 MISMATCH=1 * just turn on the mismatch MC simulation only.

- Define the number of Monte Carlo iterations to run MC simulations.

Example: .AC LIN 2 10K 100K Sweep Monte=200

For Cadence Spectre

- Copy model files into your simulation directory.

- Define sigma value for MC simulation.

Example: .param sigma=3

include "/1110ae_pcap33_v012_mc_corner.lib.scs" section=CaseName

, where CaseName could be mc, typ, max, or min.

- Define device condition by 'Xxx' statement.

Example:

Xncap (Vgg Vbb) DeviceName lf=2u wf=5u nf=4 mf=1 m=1 mis_flag=1

- m: number of devices

- mf: the same as "m"

- mis_flag: the flag parameter is used to turn on/off a mismatching parameter for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

- Define the number of Monte Carlo iterations and control MC simulations.

Example 1: run mismatch simulation only

mc1 montecarlo numruns=200 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes

Example 2: run process simulation only

mc1 montecarlo numruns=200 seed=1 variations=process donominal=yes
savefamilyplots=yes

Example 3: run process & mismatch simulation together

mc1 montecarlo numruns=200 seed=1 variations=all donominal=yes
savefamilyplots=yes

For Mentor Graphics Eldo

- Copy following files into your simulation directory.

- Define sigma value for MC simulation.

Example: .param sigma=3

.lib "/1110ae_pcap33_v012_mc_corner.lib.eldo" CaseName

, where CaseName could be mc, typ, max, or min.

- Define device condition by 'Xxx' statement.

Example:

Xncap Vgg Vbb DeviceName lf=2u wf=5u nf=4 mf=1 m=1 mis_flag=1

- m: number of devices

- mf: the same as "m"

-mis_flag: the flag parameter is used to turn on/off a mismatching parameter for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

- Define the control of Monte-Carlo simulation.

Indicator:

PROCESS: 1 to turn on the process MC simulation

0 to turn off the process MC simulation

MISMATCH: 1 to turn on the mismatch MC simulation

0 to turn off the mismatch MC simulation

Example 1:

PARAM PROCESS=1 MISMATCH=1 * turn on process and mismatch MC simulations at the same time.

Example 2:

PARAM PROCESS=1 MISMATCH=0 * just turn on the process MC simulation only.

Example 3:

.PARAM PROCESS=0 MISMATCH=1 * just turn on the mismatch MC simulation only.

- Define the number of Monte Carlo iterations to run MC simulations.

Example: .AC LIN 2 10K 100K Sweep Monte=200

ModelFileName: l110ae_pcap33_v012_mc

DeviceName: pcap_33

The PCAP model is valid within the following geometry, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed. Due to the consideration of capacitance measurement resolution, this model card is suggested not to extrapolate to the devices with the area ($lf*wf*nf*m$) smaller than $2\text{ }\mu\text{m}^2$.

Geometry description (before shrinkage):

$0.35\text{ }\mu\text{m} \leq lf \leq 2\text{ }\mu\text{m}$

$2.5\text{ }\mu\text{m} \leq wf \leq 10\text{ }\mu\text{m}$

$2 \leq nf \leq 8$

$*total\text{ area } (lf*wf*nf*m) \geq 2\text{ }\mu\text{m}^2$ (for mismatch simulation)

Bias Range:

$-3.3\text{V} \leq VGS \leq 3.3\text{V}$

$VDS = VBS = 0\text{V}$

Temperature range:

$25\text{ }^\circ\text{C}$

Layout example:

The following figure is the layout example of the PCAP. If and wf are the single poly finger length and width, respectively, and nf is the total finger number.

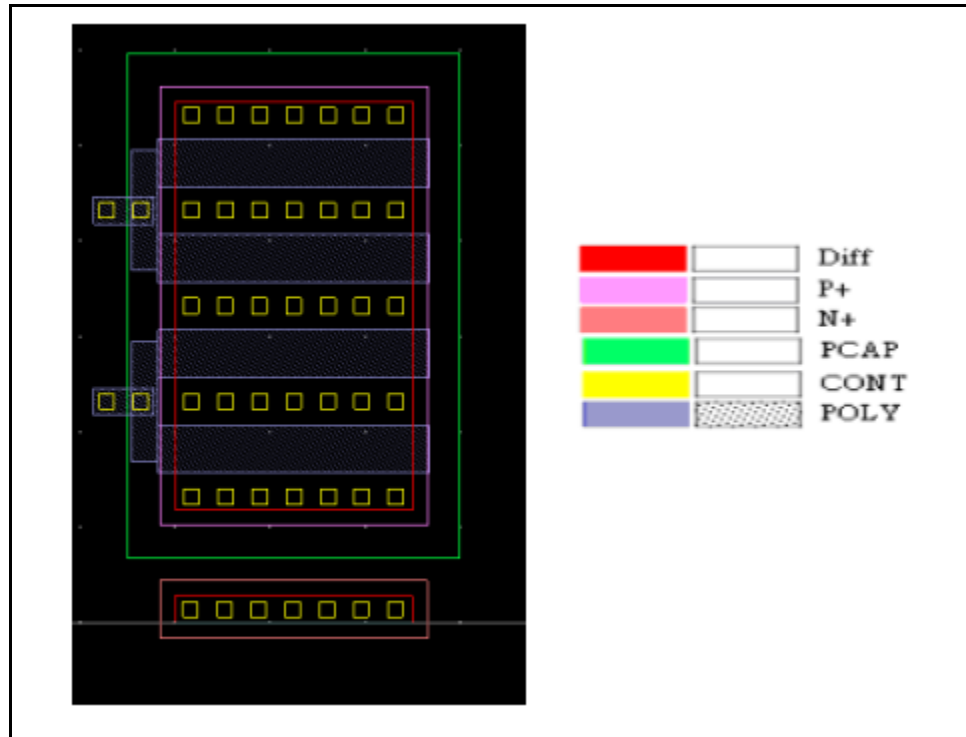


Figure 1-1 PCAP layout

2 Monte Carlo Simulation Results for Process and Mismatch Variation

Two hundred (200) Monte Carlo iterations are performed in the following simulation conditions and the following corner points show TYP, MAX, and MIN values, individually.

A reasonable number of Monte Carlo iterations is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to $(\text{iteration times})^{-1/2}$.

2.1 Simulation Result Plot For Process Variation

The following figure show out PCAP's capacitance process variation (process=1, and mismatch=0) at 25 °C :

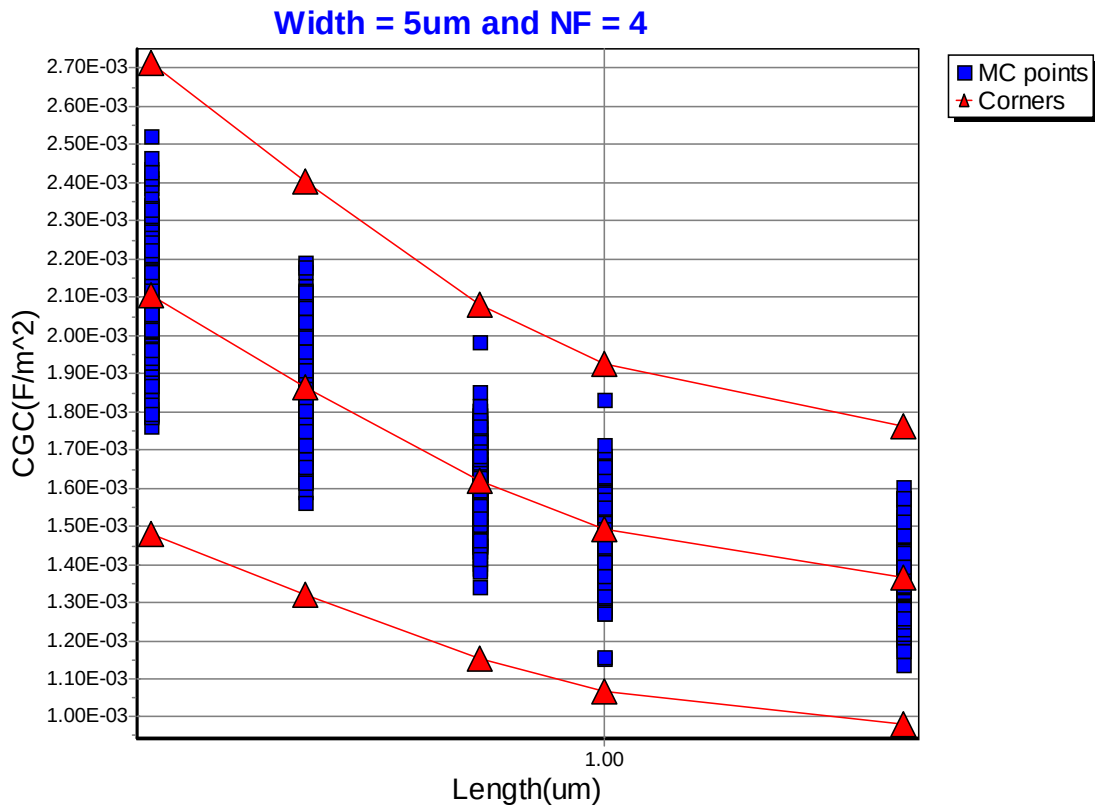


Figure 2-1 Monte Carlo plot of PCAP's capacitance

2.2 Simulation Result Plot For Mismatching Variation

The following figure show out PCAP's capacitance mismatching variation (process=0, and mismatch=1) at 25 °C :

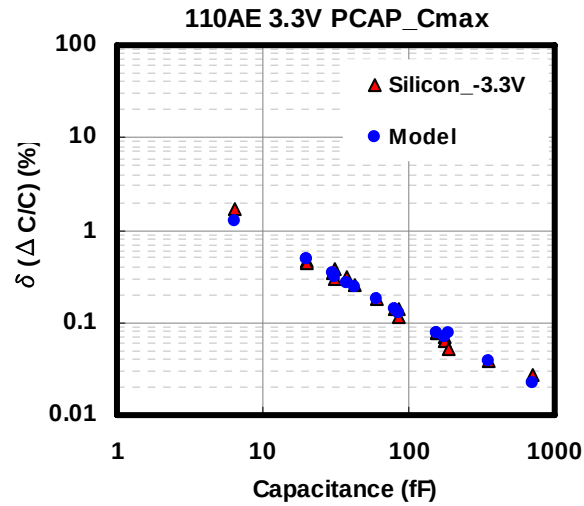


Figure 2-2 Mismatching simulation of PCAP's capacitance

3 Appendix

3.1 Hspice Warning Message

Table 3-1 Hspice warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 Eldo Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 Spectre Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1	None	None